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ROCK MAGNETIC PROPERTIES AND STRUCTURAL DEVELOPMENT IN THE CORE OF THE HAZARA-KASHMIR SYNTAXIS, NE PAKISTAN

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Abstract. Paleomagnetic and structural analyses have been conducted on three well-exposed sections through a 6-8 km thick pile of molassic red beds (Murree formation) in the lowermost tectonic unit of the Hazara-Kashmir Syntaxis. Micropaleontological age determinations of the lowermost Murree sediments indicate Late Paleocene deposition (55 m.y.). From south to north, the sections are situated in the Jhelum, Neelum (both in Azad Kashmir), and Kaghan (northeastern Pakistan) valleys. Thermal demagnetization experiments suggest that haematite with high unblocking temperatures carries stable characteristic remanence directions. The relationship between finite strain and magnetic fabric was established by mapping deformed reduction spot strain markers and by measuring the anisotropy of magnetic susceptibility (AMS). For the Jhelum valley the weakly tectonized Murree beds are characterized by flattened AMS ellipsoids resulting from diagenetic compaction. Inclination values suggest that the Murree foreland basin started to develop at about 8°N during the early suturing of India and the development of island arcs to the north. India has moved northward by at least 2600 km since collision with Eurasia in the Paleocene. Declination values suggest 45° of clockwise rotation of the block supporting the Murree formation relative to the Indian craton. For the Neelum and Kaghan valleys, quantitative strain mapping shows a progressive increase of deformation northward. NRM directions rotate passively toward the cleavage plane which parallels the foliation of the AMS ellipsoids. NRM directions and AMS ellipsoids are deformed because of superposition of tectonic strain on a primary compactional strain. The AMS pattern is interpreted in light of this superposition, and a regional deformation path from south to north is suggested. A tectonic rotation model is proposed which is consistent with the transport directions around the Hazara-Kashmir Syntaxis and the rotation of thrust sheets indicated by the NRM data. This model relates the convergent transport directions with the mean indentation direction of India into Asia.

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1. INTRODUCTION

The study area is located in the NW part of the Himalayan range in Hazara (Pakistan) and Azad Kashmir. The large-scale tectonic structure of the region is characterized by a structural reentrant or syntaxial bend, the Hazara-Kashmir Syntaxis (HKS). Because of its complex composition and key position for the geology of the northwestern Himalayas, several very detailed stratigraphic, structural, and petrographical investigations have been undertaken on the HKS [Wadia, 1931; Calkins et al., 1975; Bossart et al., 1988; Bossart and Ottiger, 1989]. This paper presents new paleomagnetic results from three sections (Jhelum, Neelum, and Kaghan valley) across the core of the HKS.

One result of previous structural investigations [Bossart et al., 1988] was the observation of increasing finite-strain state from south to north in the core of the syntaxis. This result was taken into consideration when we prepared the present study. Primary paleomagnetic directions are more likely to be found

in the southern sections than in the northern ones, where primary directions are obliterated by tectonic strain. Our paleomagnetic approach is therefore twofold. On one hand, the weakly deformed southernmost section in the Jhelum valley is useful for a regional paleogeographic and tectonic interpretation, using the primary paleomagnetic directions for determination of the paleolatitude and relative rotations of the Murree formation. This paleomagnetic information leads to a modification of our previous kinematic model for the formation of the syntaxis [Bossart et al., 1988], resulting in a more unique solution. On the other hand, the broad spectrum of natural strains preserved in the rocks gives information about progressive deformation. By measuring the anisotropy of magnetic susceptibility (AMS), a qualitative agreement of the strains observed in the field and those derived from magnetic fabrics was obtained. Finally, a deformation path from south to north is suggested, which refers to the more local deformational aspect of the syntaxis.

2. GEOLOGICAL REVIEW

2.1. *The Tectonic Structure of the Hazara-Kashmir Syntaxis*

The Hazara-Kashmir Syntaxis consists of a series of overlapping thrust sheets. They are made up of various Precambrian, Paleozoic, and Mesozoic formations which have been overthrust on Tertiary molasse deposits, known as the Murree formation. Two major thrusts can be distinguished: a higher Panjal thrust and a lower Murree thrust (Figure 1, inset). The Murree thrust corresponds to the Main Boundary Thrust (MBT) in India. Both thrusts show bowed strike trends as a result of folding and doming-up in the syntaxial bend. Along the eastern border of the syntaxis the Murree and Panjal thrust planes dip 50°-70° toward the NE and along the apex 50°-70° toward the N. Along the western border of the syntaxis the thrusts become steeper and merge together in one subvertical thrust between Balakot and Muzaffarabad.

The main lithotypes forming the Subhimalayan tectonic unit, or core of the syntaxis (footwall of Murree thrust), are sandstones and shales of the Murree formation. These deposits are interpreted as developing in an environment of meandering tidal channels in a continuously subsiding foreland basin [Bossart and Ottiger, 1989]. Age determinations based on syndepositional nummulites and assilines indicate a late Paleocene (55 m.y.) to early-middle Eocene age (45 m.y.). Metamorphism reaches prehnite-pumpellyite facies in this Subhimalayan tectonic unit.

The Murree thrust separates the Subhimalaya from the lower elements of the Lesser Himalaya, known as the Panjal imbricate zone. This zone includes imbricate slices consisting of Carboniferous to Permian tilloides, Permian metabasalts and metatuffs, Panjal volcanics, and Triassic shallow marine carbonates. A higher tectonic element of the Lesser Himalaya lies above the Panjal thrust and consists of Precambrian rocks, known as the Salkhala unit. It is exposed in the north and northeastern part of the syntaxis and consists of various crystalline rocks which have undergone upper greenschist grade metamorphism during the last main Himalayan deformation (i.e., the thrusting phase). In the hanging wall of the Salkhala unit there is a 1-5 km wide mylonite shear zone which becomes very distinct west and northwest of Balakot (Figure 1, top left). The shear direction in this region is almost parallel to a subhorizontally directed stretching lineation and is sinistral relative to the map pattern. The western part of the syntaxis has been displaced toward the south some tens of kilometers relative to the eastern part. Greco et al. [1989] have shown that this mylonite shear zone separates the tectonic units of the Lesser and Higher Himalaya and is therefore related to the Main Central Thrust (MCT).

The core of the syntaxis (Figure 1) is best interpreted with a dome model [Bossart et al., 1988]. The two limbs of the domal fold are characterized by a change in the bedding-cleavage relationship: in the Kaghan section, the bedding-cleavage intersection lineations (equals fold axes) dip steeply to the NE, North, and NW, whereas in the Neelum section they dip with intermediate angles to the SE (see stereograms of Figure 1). This domal fold in the Murree formation developed by layer compression subperpendicular to a southwestward overthrust direction. This direction was derived by finite cleavage and stretching lineations in the Murree formation (mineral reorientation parallel to the longest axis of deformed reduction spots). The Murree thrust cuts down the Murree strata from NE to SW. This out of sequence sole thrust must be, at least in part, later than the early doming, but it became involved in the later doming-up process. The hanging wall material of the Murree formation was most probably eroded and deposited in the Siwalik molasse.

2.2. *The Formation of the Hazara-Kashmir Syntaxis*

Three tectonic phases are responsible for the formation of the syntaxis [Bossart et al., 1988]. The first Himalayan phase includes overthrusting of Precambrian units (e.g., the Salkhala unit) over Late Paleozoic - Mesozoic units (the Panjal imbricate zone). Further compression caused thrusting (Murree thrust) of the previously formed tectonic elements over the Murree formation, first cutting the rising domal fold and later leading to the formation and intensification of the dome structure. A tectonic cooling fission track age on rocks near the apex of the Hazara-Kashmir syntaxis, dated at 16 m.y. by Zeitler [1985], suggests an updoming in the lower Miocene. The transport direction of all these events was southwestward. Subsequently, a progressive change of

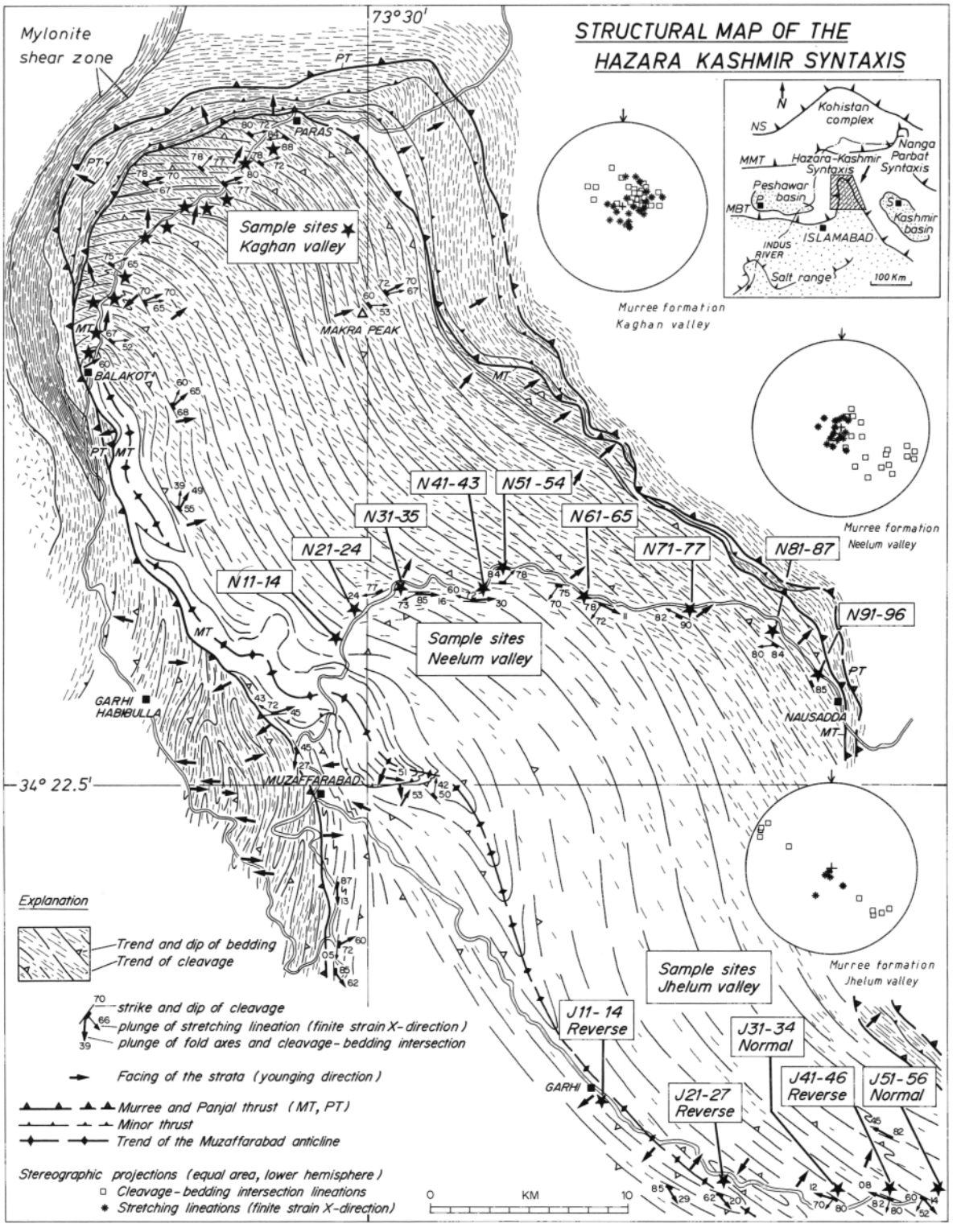


Fig. 1. Sampling sites of the Jhelum, Neelum, and Kaghan sections. In the Jhelum (J) and Neelum (N) sections the sites are numbered from west to east. The first digit indicates the site whereas the second digit denotes the sample number. Structural map based on field and survey mapping by P. Bossart, D. Dietrich, A. Greco, R. Ottiger, and J. G. Ramsay.

overthrust direction led to a southward transport direction producing the mylonite shear zone in the west of the syntaxis. The final phase resulted in a south-southeastward thrust movement recorded in the western area of the syntaxis. This mass transport is responsible for superimposed folding in the Muzaffarabad area as well as formation of a crenulation cleavage in the higher thrust units. In conclusion, a counterclockwise rotation of the overthrust shear direction from NE-SW to NNW-SSE appears to best account for the structural and geometric data we have observed.

3. NATURAL REMANENT MAGNETIZATION AND ANISOTROPY OF MAGNETIC SUSCEPTIBILITY

3.1. Evaluation of Characteristic Magnetization Components

As a result of an earlier pilot study on the natural remanent magnetization (NRM) of the Murree formation, sampling was concentrated mainly on oriented sandstone hand samples (Figure 1). The NRM of the cylindrical rock specimens drilled from the blocks (diameter 2.54 cm, height 2.2 cm), was measured using a highly sensitive ScT cryogenic magnetometer with a noise level less than 10^{-5} Am^{-1} [Goree and Fuller, 1976]. The NRM intensity averages 1.06 mAm^{-1} for the whole collection (n=251 specimens), with mean values of 1.69 mAm^{-1} (n=45) in the Jhelum valley, 0.83 mAm^{-1} (n=88) in the Neelum valley, and 1.06 mAm^{-1} (n=118) in the Kaghan profile. A significant difference in intensity between the three profiles cannot be recognized.

The NRM stability was tested in a few cases by alternating field (af) demagnetization, but usually by thermal demagnetization (Figure 2). In many samples, one or two strong secondary components with variable maximum unblocking temperatures (150°-400°C) and low coercivity (found during af cleaning, but not demonstrated here) were removed at first. At higher temperatures a stable component trending toward the projection origin of orthogonal vector plots was often observed, termed the characteristic remanent magnetization (ChRM). The ChRM resides in haematite since the maximum unblocking temperatures exceed 600°C. Its direction was evaluated using either linear regression analysis after visual inspection of the demagnetization curves or simply by averaging the stable end point vectors if the intensity changed only little during demagnetization. Mineralogical changes at elevated temperatures sometimes prevented clear definition of the ChRM.

3.1.1.The paleomagnetic section in the Jhelum valley. The ChRM directions, when corrected for tectonic tilt about the strike axis, have shallow inclinations (mean inclination of 17.2°, Figure 3). Clearly documented polarity changes favor an interpretation of the ChRM as being of early, synsedimentary primary origin. A fold test, in a classical sense [Graham, 1949], which could distinguish between a pretectonic or posttectonic magnetization age is not available. The confidence interval (α_{95}) of the dip-corrected directions, however, taking into account the observed from site to site variable attitude of the strata, is smaller than that of the in situ directions (Figure 3). Fisher's [1953] precision parameter k more than doubles from k=4.9 before tilt correction to k=9.9 after tilt correction demonstrating a pretectonic age of magnetization at the 95% confidence level [McElhinny, 1964]. The reduction of the directional scatter by this simple dip correction, which also seems to be justified by the subhorizontal bedding-cleavage intersection lineations (fold axes, see also stereogram of Figure 1: Jhelum valley) in this section, lends further support to the assumption that the ChRM is of early, depositional origin.

Despite the relatively large scatter, declination values (D=22.5°, $\Delta D_{95}=\pm 9.7°$) indicate a clockwise rotation of the Jhelum valley Murree formation relative to geographic north. Two factors may contribute to the declination scatter. The fold axes in the Jhelum valley plunge subhorizontally with plunges of up to 20°, either WNW or SSE to SE (Figure 1), thus introducing errors in the individual declinations of roughly the same order of magnitude. No attempt was made to correct for the fold axes plunge since actual plunges could not be measured at the sampling sites. Internal magnetic deformation may also deflect the remanence vectors [e.g., Kligfield et al., 1981]. The degree of deformation in this section must be weak since pressure solution is absent and the majority of AMS ellipsoids reflect a sedimentary fabric (see section 4.1.3.).

3.1.2.The paleomagnetic section in the Neelum valley. The ChRM directions of the Neelum valley show shallow to intermediate inclination values (Figure 3). No polarity changes are observed and practically all the dip-corrected remanence vectors lie in the upper hemisphere of the stereogram. The dip-corrected declinations are largely dispersed but point to a more NNW-SSE orientation when compared to the Jhelum section. A clear rotational trend, however, cannot be derived. These dip-corrected ChRM directions apparently are oriented in between the dip-corrected directions of the Jhelum and the Kaghan section.

3.1.3.The paleomagnetic section in the Kaghan valley. The remanence directions of the Kaghan valley, before and after dip correction, are much more scattered than those found in the Neelum section (Figure 3). Because of the steep dip of the bedding-cleavage intersections, the conventional bedding tilt correction could not be applied. Instead, we have corrected the in situ remanence directions by an inverse rotation of the material buckling. The amount and direction of material rotation in the Kaghan valley was deduced from studies of fibrous vein systems

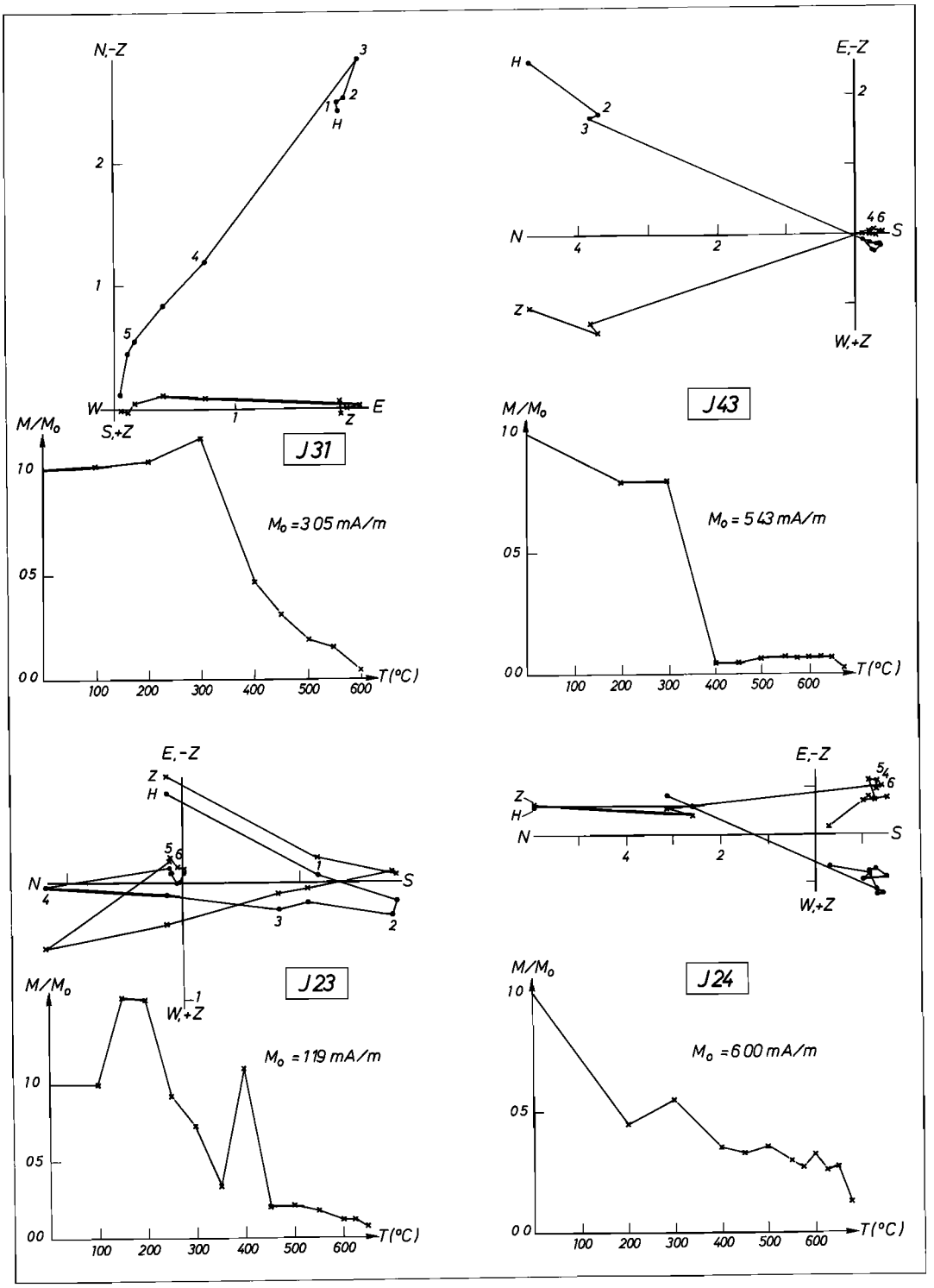


Fig. 2. Thermal demagnetization of natural remanent magnetization (NRM) of representative samples from the Jhelum section (projection of vector components after tectonic dip correction). Numbers along the horizontal (H, dots) or vertical (Z, crosses) component of the orthogonal vector diagrams indicate temperatures in Celsius divided by 100. Scale division on vector plots is mAm^{-1}.

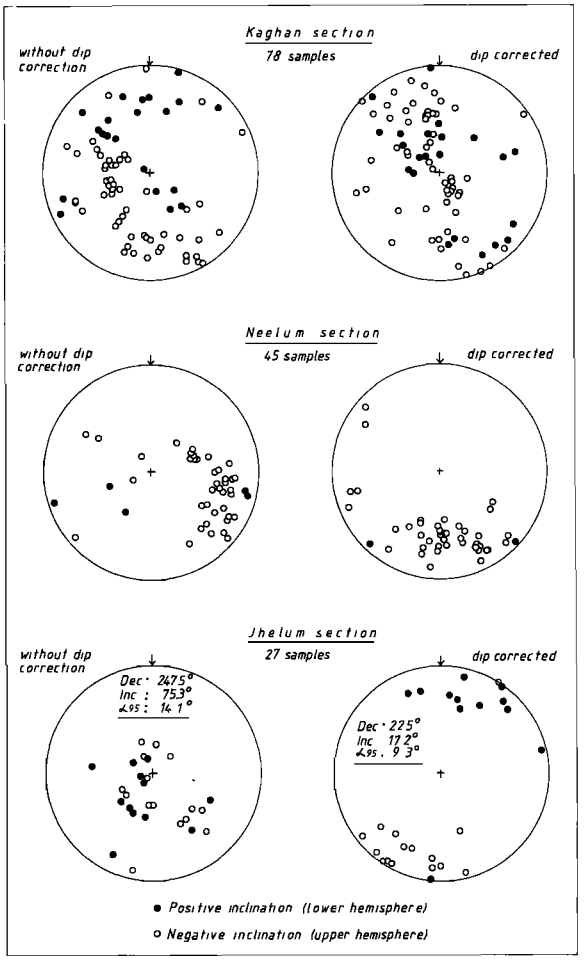


Fig. 3. Characteristic remenance directions of the Jhelum, Neelum, and Kaghan sections [open (solid) circles: upper (lower) hemisphere with negative (positive) inclinations]. Stereograms to the left indicate in situ remanence directions whereas stereograms to the right show remanence directions after tectonic dip correction. Note the increase of scatter of dip corrected paleomagnetic vectors from the Neelum to the Kaghan section.

[Bossart et al., 1988]. Material in the XY strain plane is rotated to the northwest while in the XZ strain plane there is material buckling to the southwest and northeast. A rotation of the same amount in the opposite sense is necessary to cancel the material buckling. This inverse rotation was done in the XY and XZ principal planes. As a result we have found a northwest-southeast striking girdle of dip-corrected remanence directions (Figure 3, top right).

3.2. *Anisotropy of Magnetic Susceptibility*

Magnetic minerals in the Murree formation appear to be clay minerals and micas and/or haematite. Samples for AMS measurements were chosen among the bulk paleomagnetic specimens (Figure 1). The bulk susceptibility along the sample cylinder axis was measured with a Kappa bridge (KLY-1 [Jelinek, 1973]), whereas the low field susceptibility differences in three orthogonal planes of a sample were determined using a modified Digico spinner magnetometer [Schultz-Krutisch and Heller, 1985]. The observed susceptibility differences were combined with the bulk susceptibility to calculate the directions of the principal axes (k_{min}, k_{int}, k_{max}) and the shape of the AMS magnitude ellipsoid using the Jacobi iteration method [Scriba and Heller, 1978].

Comparing the in situ and dip-corrected orientations of the principal anisotropy axes in the Jhelum section, most of the k_{min} axes are brought to a subvertical position normal to the bedding plane by the dip correction (Figures 4a and 4b). The dip-corrected k_{int} axes are widely dispersed but are shallower oriented than the correspondent in situ directions. The in situ and dip- corrected k_{max} axes are concentrated in the NW and SE quadrants. Dip correction rotates most of the k_{max} axes into nearly horizontal planes. The patterns of the principal axes in the Neelum and Kaghan sections show a clear magnetic foliation defined by flat in situ k_{int} and steep in situ k_{max} axes. This steeply dipping magnetic foliation strikes in a NW-SE and WNW-ESE direction, respectively. Most of the k_{min} axes lie normal to these magnetic foliation planes.

The shapes of the individual susceptibility ellipsoids are given by susceptibility plots (very similar to the Flinn diagrams [Flinn, 1962] used in structural geology), with the k_1/k_2 values as the ordinate and the k_2/k_3 values as the abscissa (Figure 5). In the Jhelum section, most AMS ellipsoids plot in the apparent flattening field, with a few exceptional samples plotting in the field of constriction. In the Neelum section the siltstones fall into the apparent constriction field, whereas the sandstones have been flattened in the eastern part but are constricted in the western part of section. In the Kaghan section the AMS ellipsoids of the sandstones show the highest apparent constriction values in the east, whereas westward the AMS ellipsoids become less constricted. Near Balakot the sandstones and the few siltstones measured plot in the apparent flattening field (see also Figure 11).

4. INTERPRETATION OF MAGNETIC MEASUREMENTS

4.1. *Jhelum Section*

4.1.1.Inclination of the Jhelum section. The mean deposition age of the Murree formation was determined as 50 m.y. [Bossart and Ottiger, 1989]. The 50 m.y. paleolatitude of Murree deposition, calculated from the dip-corrected mean inclination of the Jhelum sites (Figure 3), indicates a paleolatitude of

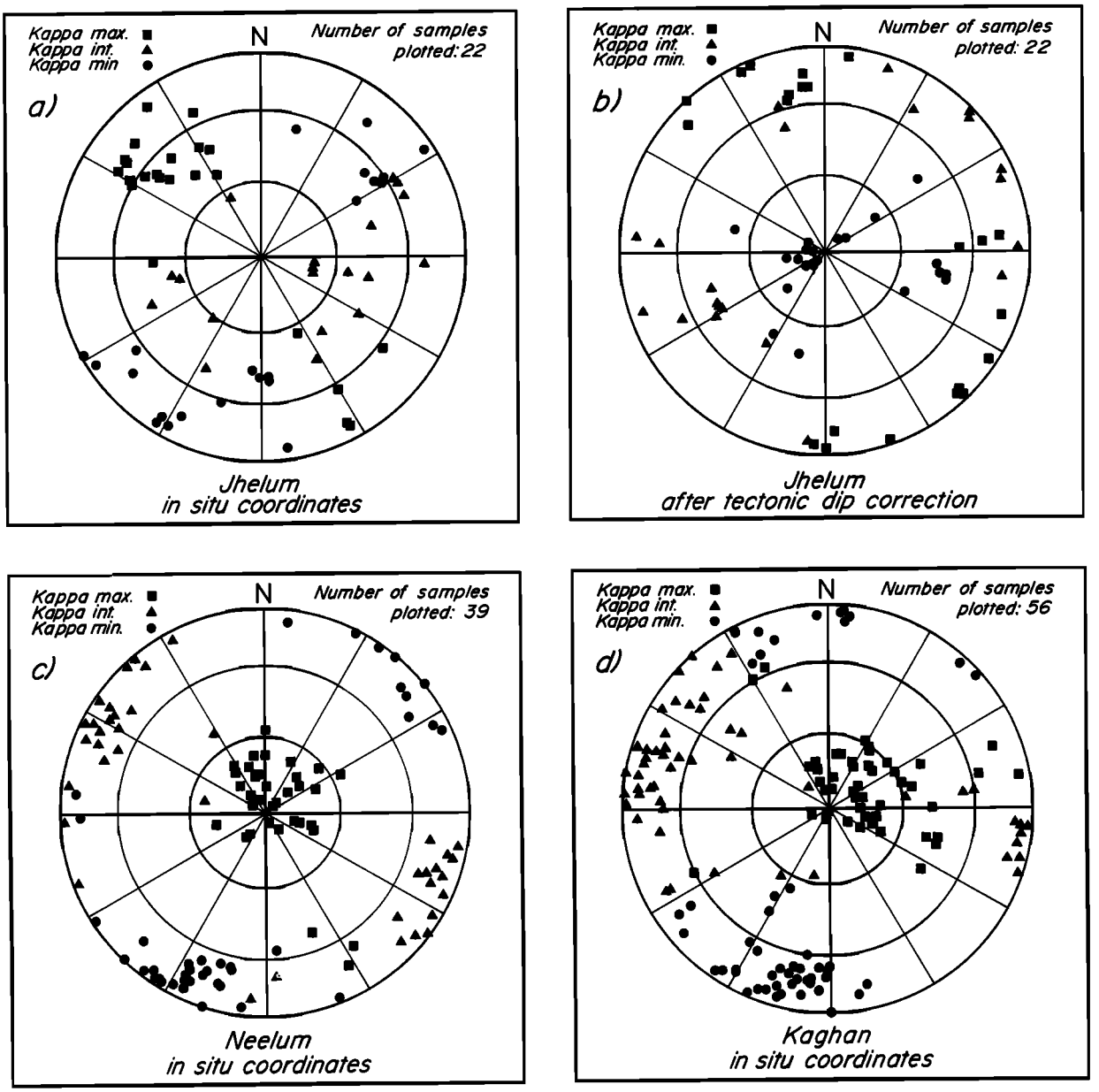


Fig. 4. Stereographic projections (lower hemisphere) of the orientation of the AMS ellipsoids (squares, maximum axes; triangles, intermediate axes; circles, minimum axes) in the Jhelum, Neelum, and Kaghan sections.

about 8° north. The original deposition locus was closer to the northwestern border of the Indian craton than its corresponding distance today because the coverblock supporting the Murree formation was detached from the basement in the lower-middle Miocene. The difference between the 50 m.y. paleolatitude and present-day latitude indicates that the Murree beds have been transported on the Indian craton at least 2600 km to the north. Several workers have demonstrated that there was a sudden decrease in the "rapid northward flight" of India between 60 and 40 m.y. [e.g., Besse et al., 1984; Patriat and Achache, 1984]. There are not enough accurate inclination data in the Jhelum section for demonstrating this speed decrease which is clearly shown by the magnetic anomalies of the Indian Ocean [e.g., Powell, 1979].

4.1.2.Declination of the Jhelum section. The mean declination of the dip-corrected ChRM (D= 22.5°±9.7°) in the Jhelum valley indicates a clock-

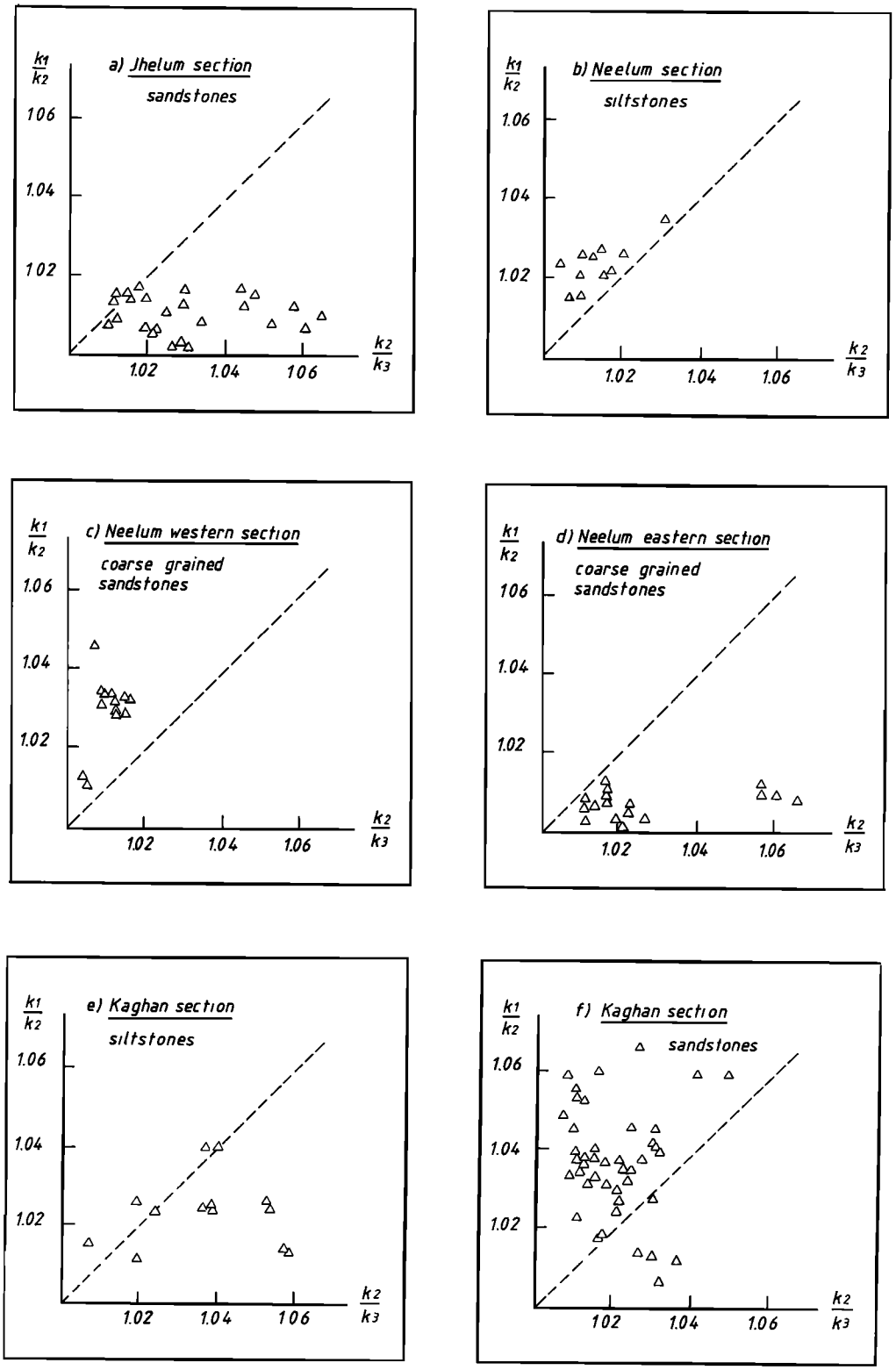


Fig. 5. Susceptibility plots of the AMS ellipsoids from the Jhelum, Neelum, and Kaghan sections (k_1, maximum axes; k_2, intermediate axes; k_3, minimum axes of the AMS ellipsoids). Values below the 45° line represent the field of apparent flattening, while values above represent the field of apparent constriction. (a) Jhelum section, (b)-(d) Neelum section (western and eastern part), (e) and (f) Kaghan section.

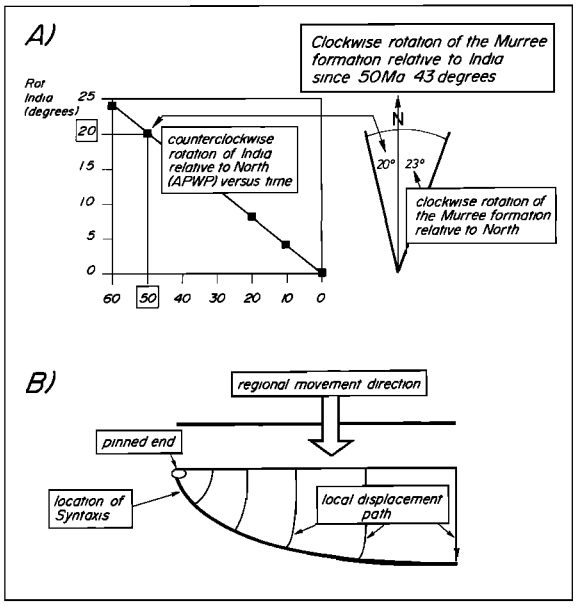


Fig. 6. (A) The block supporting the sediments of the Murree formation has rotated 43° in a clockwise sense relative to the Indian craton since 50 m.y. (B) A simple bending model consistent with the clockwise rotation of the Murree formation relative to India: a regional movement or transport direction with a pinned end creates divergent lineation patterns. In the region of the Hazara-Kashmir Syntaxis, transport directions swing clockwise away from their original orientation, as indicated by the local displacement path.

wise rotation of the block underlying the Murree Formation since the time of deposition (Figure 3). In order to calculate the amount of rotation for the Murree beds relative to the stable Indian craton, counterclockwise rotation of India since 50 m.y., deduced from the Indian apparent polar wander path (APWP), has to be added. The Tertiary segment of the Indian APWP is not well constrained by paleomagnetic data from autochthonous rock units [Klootwijk, 1979]. Using ocean floor magnetic anomalies between India and Madagascar [Powell, 1979] and data from several hot spots [e.g., Patriat and Achache, 1984], Klootwijk et al. [1985] have constructed a well-defined Tertiary APWP curve for India which is in agreement with the relative movements of the other Gondwana continents. On this basis, a counterclockwise rotation of 20° of India relative to geographic north since 50 m.y. has been postulated. The clockwise rotation of the Murree beds relative to India is then calculated as 20° + 23° = 43° (Figure 6A). This amount of clockwise rotation (estimated error is ±10°) can explain an aspect of the curvature in the Western Himalayas: if one assumes a pinned end in the NW of the syntaxis, a southward directed regional movement swings local transport directions clockwise away from their original north-south orientation (Figure 6B). This divergence of transport directions has also been confirmed by structural mapping [Mattauer, 1986; Coward et al., 1987; Bossart et al., 1988]. West of the pinned end the regional movement directions should be parallel to local transport directions.

4.1.3.AMS of the Jhelum section. In the southernmost profile, practically no tectonic deformation (cleavage) is visible in the rocks sampled, that is, the coarse-grained sandstones. This is supported by the orientation and shape of the AMS ellipsoids: most of the tilt-corrected k_3 axes are generally normal to the horizontal bedding plane and consequently most of the k_1 and k_2 axes girdle on a very shallow great circle which represents a subhorizontal major susceptibility plane. In the susceptibility plot (Figure 5) practically all of the data points lie within the field of apparent flattening. Both the orientation of the k_3 axes and the shape of the ellipsoids are taken to reflect sedimentary compaction. The maximum axes (k_1) cluster around a flat lying NNW-SSE direction that can be interpreted as a direction of depositional current flow.

Nearly one quarter of the dip-corrected k_3 axes deviate by more than 45° from the vertical (Figure 4, top left). The AMS ellipsoid shapes of these samples plot just in or near the field of apparent constriction (Figure 5a). An incipient tectonic strain regime seems to be indicated resulting in a "pencil structure stage" [Graham, 1978; Kligfield et al., 1981]. This mild tectonic imprint, however, is restricted to local importance in the Jhelum valley.

4.2. *Neelum Section*

4.2.1.NRM directions of the Neelum section. In the Neelum section two regional situations with uniform rotation patterns of the NRM directions are met (Figures 7A and 7B): in the eastern part a clockwise rotation of in situ remanence directions (N61-64 to N91-96, Figure 7A) and in the west a counterclockwise rotation of dip-corrected remanence directions (N11-14 to N51-54, Figure 7B).

Structural mapping has shown that the cleavage intensity increases from a very mild pressure solution cleavage in the west to a strong slaty cleavage in the east (Figure 7). Furthermore the cleavage strike changes through the section from a W-E orientation in the west to a NNW-SSE orientation in the east (Figure 1). Therefore the increase in strain and cleavage intensity in the eastern part of the section rotates the remanence vectors into a subparallel orientation to the direction of the planar cleavage fabric. The obvious clockwise rotational trend of cleavage and in situ remanence directions suggests a passive rotation of the characteristic remanence directions rather than an overprint of secondary directions.

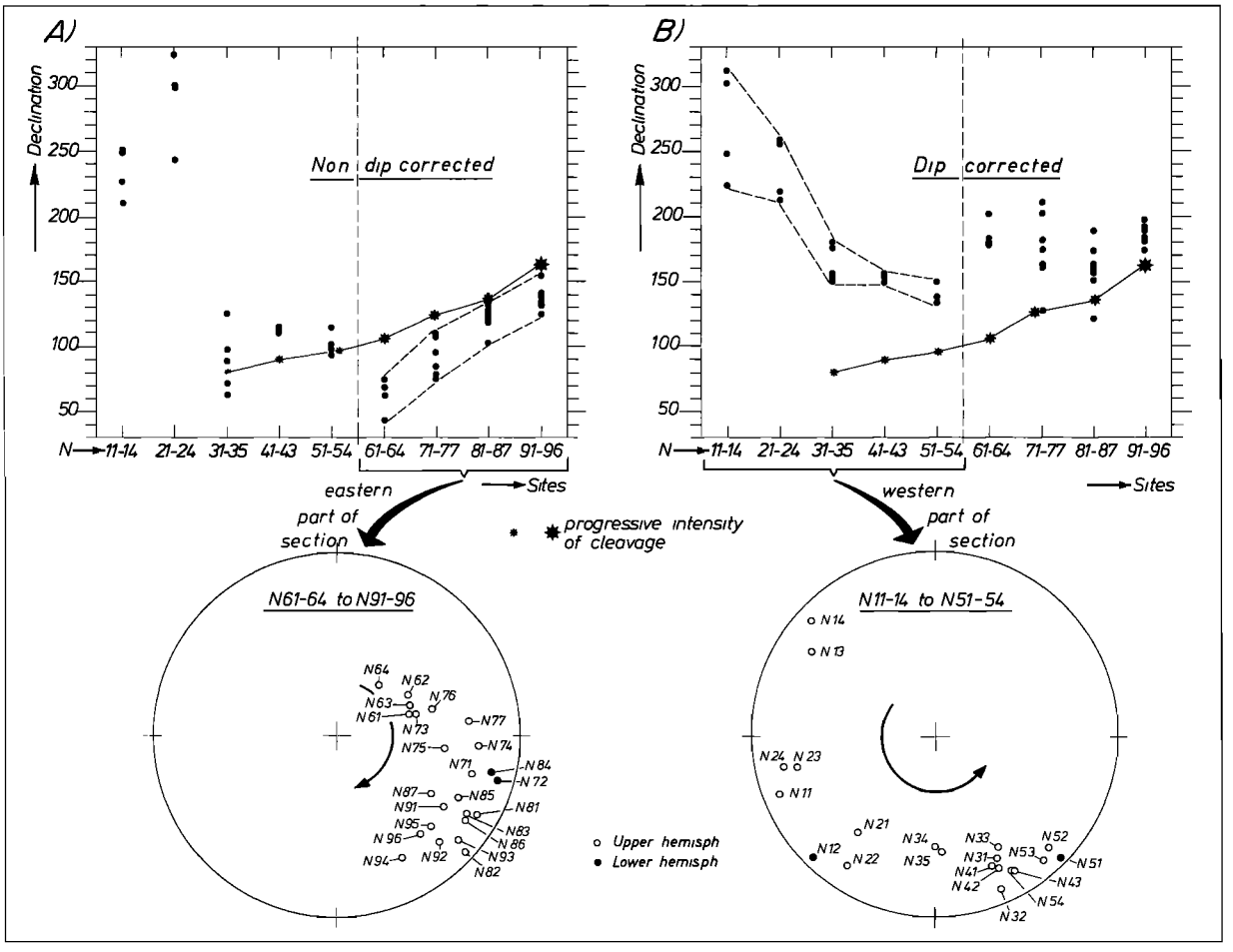


Fig. 7. (A) Declination trend of in situ remanence directions in the Neelum valley with stereographic projection of samples in the eastern part of section (N61-64 to N91-96). For location see Figure 1. (B) Declination trend of dip-corrected remanence directions with stereographic projection in the western part of section (N11-14 to N51-54). The finite pressure solution cleavage (strike indicated by stars) increases from west to east.

The strongly dispersed remanence directions in the western part of the section are interpreted as being affected by the superimposed folds recognized in the Muzaffarabad area [Calkins, 1966; Bossart et al., 1988]. The base of the Murree formation, the Muzaffarabad anticlinal structure, was refolded with SSE directed compression axes during the late deformation phase. In particular, the dip-corrected characteristic directions of the westernmost samples (N11-14, Figure 7B) are considerably scattered because of individual passive reorientation during refolding. Toward the east (N41-43 to N51-54) the increasing cleavage intensity starts to effect the samples. Between these two areas (N21-24 to N31-35), where both effects are small, the directions conform with the dip-corrected primary directions observed in the Jhelum valley (compare stereograms of Figure 7B and Figure 3, bottom right).

Hence only a very small portion of the Neelum section contains primary remanence directions. The majority of the samples have been influenced by local deformation and cannot be used for a regional tectonic interpretation.

4.2.2.AMS of Neelum section. The orientation and shape of the AMS ellipsoids mark the advancing strain state of the rocks. The orientation of principal axes is constant throughout the section (Figure 4). The siltstones all lie in the apparent constricton field. The coarse-grained sandstones show different patterns. Prolate AMS ellipsoids in the western part of section are distinguished from oblate ones in the eastern part (Figure 5). Differences in the local deformation style in the west with respective to the east may be responsible for this strain pattern. The late NNW-SSE directed deformation phase superimposed onto the early NE-SW directed one (the strains of the

early phase are only weakly recorded in the west) probably has "overprinted" the originally oblate ellipsoids, resulting in these prolate cigar-shaped ellipsoids. In the east an intense finite-pressure solution cleavage is the result of the early deformation. The flattened AMS ellipsoids may therefore record increasing pressure solution fabric in the eastern part of the section. Either haematite or other flaky minerals, such as clay minerals and micas, may have been reorientated initially, with their basal plane parallel to the pressure solution cleavage. A flattened mineral shape may have been augmented by solution and crystal growth under low-grade metamorphic conditions.

4.3. Kaghan Section

4.3.1.NRM directions of the Kaghan section. The finite strain increases from the Neelum valley to the Kaghan section. The initially flat oriented intersection lineations (fold axes) successively align parallel to the steeply NE dipping mineral stretching lineations (Figure 1). In addition, a strong NE-SW striking slaty cleavage is developed in the Kaghan section due to pressure solution.

Most of the in situ ChRM directions cluster near to a NW-SE striking, steeply to the NE dipping plane which is parallel to the cleavage plane (compare ChRM directions in Figure 3 with the cleavage strike trend in Figure 1). This is interpreted as passive rotation of ChRM directions toward the cleavage plane during formation of the pressure solution cleavage. The girdle of dip-corrected ChRM directions also reflects the overall strike trend of this slaty cleavage. Even our modified tectonic dip correction (inverse material rotation) does not produce the same directional pattern as that found in our reference profile in the Jhelum valley. Additional inverse strain correction (i.e., the strain amount producing the cleavage) would be necessary. Furthermore, no systematic alignment of the magnetic vectors parallel to the stretching lineations has been observed from the Neelum to the Kaghan valleys. The magnetic vectors are randomly oriented within the cleavage planes.

4.3.2.AMS of Kaghan section. The shape of reduction spots within the clayey silts of the Murree formation yields strain ellipsoids of the apparent constricted type in the east and of the apparent flattened type in the western Kaghan section. The constrictional ellipsoids can be explained by the doming-up process of the syntaxis apex region [Bossart et al., 1988]. The flattened ellipsoids can be attributed to the influence of the huge shear zone (MCT) passing west of Balakot.

The shape and orientation of the deformed reduction spots and of the AMS ellipsoids generally conform qualitatively, supporting the aforementioned structural interpretation. The maximum AMS axes plunge steeply toward the NE (Figure 4d) and are parallel to the tectonic stretching lineations (Figure 8). The k_{max}-k_{int} planes are parallel to the regional cleavage trend (XY plane of reduction spots). Most of the silt and fine-grained sandstone AMS data plot in the apparent flattening field (Figure 5e). Only the few samples from the eastern part of the profile are constricted. The more competent coarse-grained sandstones are confined mostly to the field of apparent

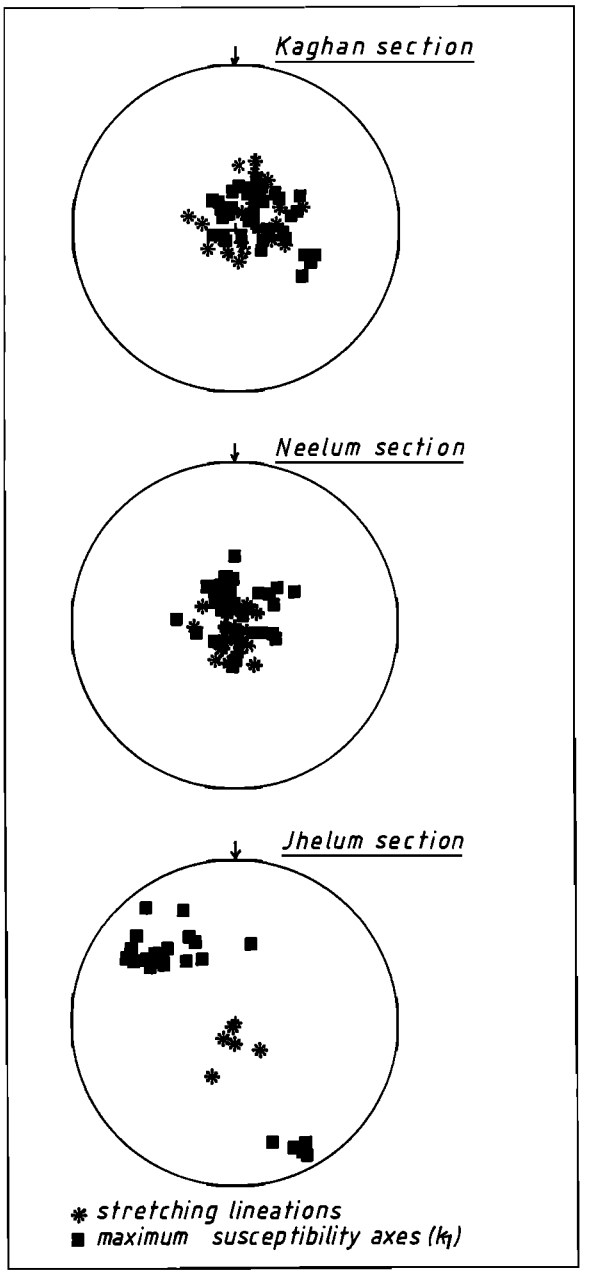


Fig. 8. Stretching lineations (maximum extension directions of the finite strain ellipsoids derived from deformed reduction spots) and in situ maximum axes of the AMS ellipsoids (k_1) from the Jhelum, Neelum, and Kaghan sections. Lower hemisphere projection.

constriction (Figure 5f). The intensity of deformation is stronger than that of the Neelum profile . Higher k_1/k_2 values lead to a prolate shape of the Kaghan ellipsoids. The few data points in the apparent flattening field are from samples collected at the western part of the Kaghan section, close to the N-S trending mylonite shear zone.

5. DISCUSSION

5.1. A Magneto-Structural Model for the Kinematic Formation of the Syntaxis

Several models have been put forward to explain the curvature of the Himalayan chains. One regional model [Klootwijk et al., 1985] proposes counterclockwise rotational underthrusting of Greater India and consequent oroclinal bending of the Himalayan chains, with back arc spreading on the Tibetan Plateau.

Curvature of the western Himalayan chain can also be explained by our bending model (Figure 6B) where a regional southward directed mass transport swings local transport directions clockwise away toward a NE-SW directed orientation. This is supported by our small and large scale structural analysis in the core of the HKS (stretching lineations, cleavage patterns, and branch thrust lines) which suggests NE-SW directed mass transport. This direction lies normal to the overall structural trend of the HKS.

The bending model suggests limited continental underthrusting and little shortening of the hanging wall block in the vicinity of the pinned end. This pinned end is not geologically documented at the surface in the Western Himalayas, but we assume that the HKS is closer to it than localities further to the southwest or southeast. Furthermore, the bending model requires arc-radial and arc-parallel extension. There is evidence for some extension along the Himalayan arc. Tapponnier et al. [1981] indicate normal faults oriented radial to the arc. On the other hand, the mapping of deformed reduction spots in the syntaxis gives no indication of such extensional strains. A further disadvantage of this model is the mass compatibility; huge block rotations around the pinned end would cause mass incoherency (holes and material accumulation) which is not supported by seismic data [Seeber et al., 1981]. Therefore the term 'block rotation' is correct in the sense that only the cover, or part of the cover, has rotated. The deeply buried basement in the HKS probably would indicate different rotation directions and amounts.

To prove the consistency of the bending model described above (Figure 6B), rotation of tectonic blocks relative to the Indian craton on a larger scale has to be studied. Therefore the available data of Tertiary NRM measurements beyond the Main Boundary Thrust in Northern Pakistan and Indian Kashmir were compiled (Table 1). Three main features can be seen from the rotation of different lithological units relative to the Indian craton (Figure 9):

1. Independent of an early or late Tertiary age of the formations, the magnetic vectors of crustal blocks supporting Tertiary lithological units tend to lie normal to the structural trend (folds, thrusts).

2. In the core of the syntaxis and southeast of it (localities A, C, and D, Figure 9), a clockwise rotation of the Murree formation relative to the Indian craton of about 45° is observed. Locality B is an exception explained by a local structural "disturbance" [Klootwijk et al., 1986a] and can therefore be neglected in a regional interpretation. In the west of the syntaxis, both no rotation and counterclockwise rotation near the syntaxis is recorded. The amount of rotation is generally less than 30°.

3. Two different transport directions around the syntaxis were mapped [Bossart et al., 1988]. The eastern and earlier one is parallel to the clockwise rotated magnetic vectors in the syntaxis. The western and later one tends to be parallel to the slightly counterclockwise rotated magnetic vectors on the Potwar Plateau and in the Salt range region.

In the light of these observations, two questions arise with regard to the formation of the syntaxis: (1) is there a visible surface expression of the transition from "large clockwise rotations in the syntaxis" to "slight counterclockwise rotations" in the west, and (2) what is the geological reason for these changes in rotation?

We propose that the border of clockwise and counterclockwise rotations lies close and parallel to the strike trend of the Muzaffarabad anticline, merging into the mylonite shear zone at the western side of the syntaxis (Figure 1). The Muzaffarabad anticline probably masks a major blind thrust in the footwall block of the Murree formation that is documented farther to the southeast, in Indian Kashmir, where the Murree formation is thrust over Siwalik molasse deposits. The mylonite shear zone cuts the NW-SE directed structural trend of the syntaxis. Furthermore, it marks the transition zone of the early SW, to late SSE, transport directions. This discontinuity is the most obvious geological argument for the rotational changes of the magnetic vectors. To the north the border of rotational change is not clear. The increases in deformation and metamorphism distort the information contained in the primary remanence directions.

The cause of the 'block rotations' to the west of the HKS might be explained by the "salt-no salt" model of Seeber et al., [1981]. In the syntaxis and east of it there is a strong coupling of Indian basement and cover, whereas to the southwest of it the widespread and often thick Cambrian evaporite deposits act as ideal décollement horizons between basement and cover (Salt range). Large clockwise rotations, basement thrusting and thrusting at the surface occur simultaneously where evaporites are absent. There is little or no rotation where thrusting occurs in the salt.

TABLE 1. Primary Remanence Directions and Relative Rotations of Tertiary Formations in Northern Pakistan

Site Figure 9	Formation	Locality and Geographic Coordinates	Age, m.y.	Declination, deg	Inclination, deg	k-Fisher	α_{95}, deg	Samples	Rotation India APWP, deg	Rotation relative to India, deg	Reference
A	Murree formation	Jhelum valley 73.60°E/34.20°N	55-45	22.5	+17.2	10	9.3	N 12 [a] R 15	20	43 clockwise	this paper
B	Basal Murrees	Kalakot inlier ?73.85°E/33.34°N	? Early Eocene	129	-12.5	40	3.5	R 39	14-18	33-37 anticlockwise	Klootwijk et al. [1986a]
C	Lower upper Murrees	Jammu foothills ?74.30°E/33.12°N	? Eocene-Oligocene	204.5	-23	20	3.5	R 91	10-14	35-39 clockwise	Klootwijk et al. [1986a]
D	Lower Murrees	Jammu foothills ?74.30°E/33.12°N	? Oligo.-Miocene	23.5	+39	13	4.5	N 51 R 35	8-12	22-36 clockwise	Klootwijk et al. [1986a]
E	Lockhart Limestone	Chichali gorge 71.42°E/33.00°N	60-65	335	-13	11	8.5	N 28	24-26	±0	Klootwijk et al. [1986b]
F	Hangu formation	Dhak Pass 71.82°E/32.70°N	60-65	173	- 2.5	5	14	N 7 R 19	24-26	31-33 clockwise	Klootwijk et al. [1986b]
G	Upper Siwaliks	Dhaud Khel 71.65°E/32.90°N	2-5	356.6	+35.9	42.2	8.6	N 4 R 4	1-2	1-2 anticlockwise	Opdyke et al. [1982]
H	Lower - middle Siwaliks	Nagri 72.45°E/32.75°N	5-12	350.9	+34.8	18.4	5.3	N 26 R 16	2-5	4-7 anticlockwise	Opdyke et al. [1982]
I	Lower - middle Siwaliks	Chakwal Bhaun 72.75°E/32.85°N	5-12	330.7	+31.8	13.1	6.4	N 25 R 16	2-5	24-27 anticlockwise	Opdyke et al. [1982]
J	Lower - middle Siwaliks	Dhala Nala 73.20°E/32.85°N	5-12	328.5	+35.9	37.6	5.5	N 14 R 5	2-5	26-29 anticlockwise	Opdyke et al. [1982]
K	Upper Siwaliks	Kotal Kund 73.30°E/32.75°N	2-5	320.8	+34.1	15.6	4.6	N 37 R 29	1-2	37-38 anticlockwise	Opdyke et al. [1982]
L	Upper Siwaliks	Chambal Ridge 73.45°E/32.70°N	2	351.7	+37.4	17.8	3.7	N 49 R 38	1	7 anticlockwise	Opdyke et al. [1982]
M	Upper Siwaliks	Tatrot 73.35°E/32.85°N	2-5	327.7	+36.7	17.3	3.9	N 45 R 36	1-2	30-31 anticlockwise	Opdyke et al. [1982]
N	Lower - middle Siwaliks	Bora Kas 72.61°E/33.23°N	7-12.5	350.3	+36.3	9.7	7.1	N 30 R 15	3-5	5-7 anticlockwise	Tauxe et al. [1982]
O	Middle Siwaliks	Malawalla Kas 72.64°E/33.25°N	7-8.5	353.7	+35.5	8.0	9.1	N 21 R 14	3	3 anticlockwise	Tauxe et al. [1982]
P	"Foredeep Sediments"	Soan Syncline 73.10°E/33.40°N	9	317	—	—	—	—	4	39 anticlockwise	Burbank [1984]
Q	"Peshawar basin Sediments"	Dag, Garhi Chanda 73.10°E/33.40°N	2.5	358	—	—	—	—	1	0-1	Burbank [1985]

a) N and R refer to number of samples with normal and reversed polarity, respectively.

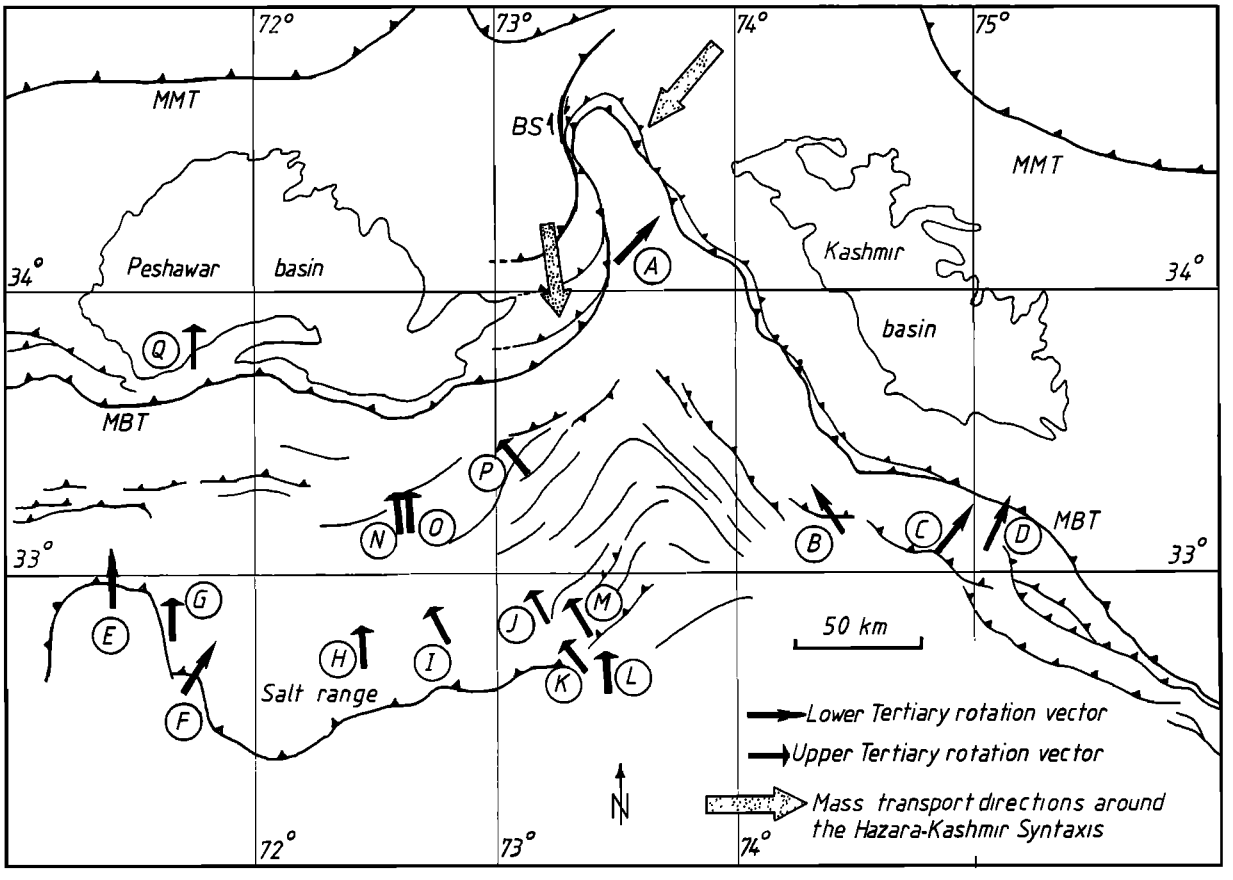


Fig. 9. Different paleomagnetic sampling localities (A-P, Table 1) with their paleomagnetically derived rotations relative to the Indian craton below the Main Boundary thrust (locality Q is above the MBT). Mainly involved are sediments of the early (60-40 m.y., long arrows) and late (<15 m.y. "Siwaliks", short arrows) foreland basin in Northern Pakistan and Indian Kashmir. Locality A lies in the Jhelum valley and is the result of the present investigation. MMT, Main Mantle thrust; BS, Balakot mylonite shear zone.

Slightly counterclockwise rotation in the Salt range and Potwar Plateau has occurred at different times in response to segmentation of the hanging wall thrust sheet [Opdyke et al., 1982]. Cross- section balancing suggests local rotations of thrust sheets around nearby poles rather than a regional counterclockwise rotation of the Salt range hanging wall block [Lillie et al., 1987; Baker et al., 1988]. Large clockwise rotations in the HKS and southeast of it belong to a large regional rotation system rather than to local rotations of thrust sheets. Klootwijk et al. [1985] suggest that the rotation center of this system lies about 70 km northwest of the HKS, at the intersection in the Indus Kohistan Seismic Zone and the Main Mantle thrust.

The paleomagnetic studies require a large clockwise rotation of the thrust sheets relative to the Indian craton in and east of the HKS, whereas west of it there is, in general, no or small counterclockwise rotation of thrust sheets relative to the craton (Figure 10A). From the mapping of the structures the formation of the HKS can best be interpreted by a counterclockwise rotation of transport directions from 220° to 160°, where the reference frame is geographic north (Figure 10B). A differentiated interpretation of the HKS can be achieved by a combination of both the magnetic studies and the structural mapping. The early transport direction (Figure 10C: from 1 to 2) was initially reversed to the mean movement vector of India and was then successively rotated clockwise to its present position (220°). The late transport direction (Figure 10C: from 2 to 3) in the west was slightly rotated counterclockwise to its present position (160°) together with the slight counterclockwise rotation of the Indian craton. It essentially never changed its orientation and was always more or less opposite to the mean Indian movement. This model relates the complicated orientation pattern of transport directions

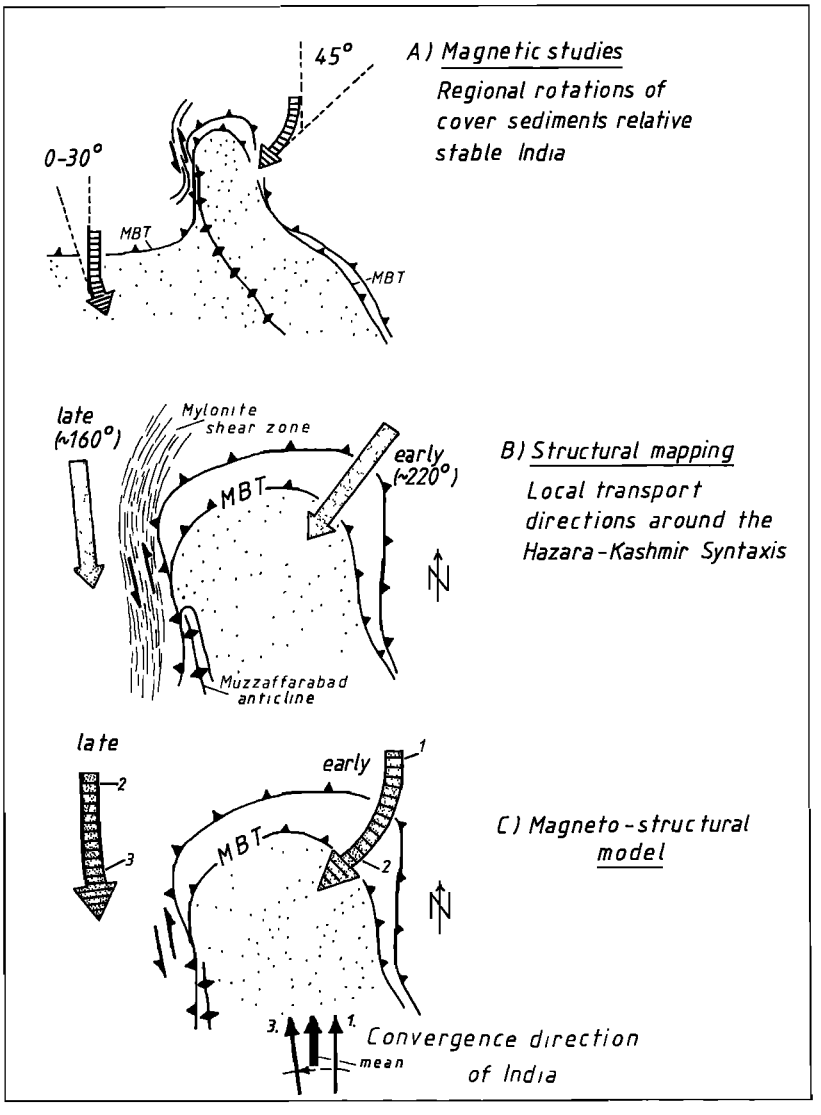


Fig. 10. A magnetostructural model (C) for the formation of the Hazara-Kashmir Syntaxis derived from a combination of the (A) regional model and the (B) local model. An early transport initially opposite to the motion of India was successively rotated to its present position (220°). The late transport direction in the west rotated little and was always approximately opposite to the movement of India.

around the HKS with the mean convergence direction of India with Eurasia. Slight counterclockwise rotation of India is indicated by increments 1 and 3 (Figure 10C). These increments can be compared with the synchronous correspondent increments of the transport directions (1 and 3). As far as the structural model, based on field mapping, is concerned [Bossart et al., 1988], we have to assume passive clockwise rotation of linear and planar deformation fabrics (e.g., stretching lineations and cleavage planes), although there is no direct field evidence for such rotations.

5.2. *A Possible Deformation Path in the Murree Formation Derived by AMS Ellipsoids*

AMS and reduction spot ellipsoids can be compared qualitatively. In doing so we assume that the magnetic mineralogy responsible for the AMS does not change substantially throughout the region investigated. This assumption is supported by similar bulk susceptibility values in the three study sections.

The south to north increase in finite strain intensity coincides with a remarkable change of AMS ellipsoid deformation, as measured in coarse-grained

sandstones (Figure 11a). In the southernmost Jhelum section (A), the AMS ellipsoids are flattened, and the bedding tilt correction brings many of the k_3 axes to a subvertical orientation denoting a preserved sedimentary compaction fabric. The only important tectonic imprint is a rigid bedding rotation with little or no internal deformation. Therefore the Jhelum valley AMS ellipsoids were chosen as the starting point for a possible deformation path in the two northern sections. The most obvious way to deduce a deformation path is to consider localities which have preserved only one main tectonic deformation phase. For our path these localities are the eastern Neelum valley section (B), followed by the more intensely deformed eastern Kaghan valley section (C), where the early deformation phase with NE-SW directed compression axes is seen. We therefore propose a deformation path from the Jhelum valley (A) to the eastern Neelum section (B) and finally to the eastern Kaghan (C) section. The western parts of the Neelum and Kaghan sections (D and E), which show two different tectonic deformation phases of different intensity, must considered separately. The AMS fabric in the western Neelum section could be directly derived by a superposition of tectonic strains onto the sedimentary fabric in the Jhelum valley. The AMS fabric in the western Kaghan section can be explained by the effect of the sinistral mylonite shear zone onto the AMS ellipsoid in the eastern section of the Kaghan valley.

Coaxial deformation, as suggested by Graham [1978] and Kligfield et al. [1981] in the Maritime Alps in France is inappropriate for the deformation history of the Murree formation. A combination of rigid body rotations and/or heterogeneous simple shear must be assumed if the dip-corrected sedimentary AMS axes in the Jhelum valley (A) are to be transformed into tectonically deformed ones in eastern Neelum (B) (Figure 11a). The resulting deformation path is illustrated in Figure 11b (left-hand side) in more detail: The Jhelum (A) sedimentary fabric can be transformed to the deformed East-Neelum (B) fabric by applying subvertical shear. This causes a deflection at the k_2/k_3 axis of the deformation path in the Flinn diagram, where the steep k_{min} axes in the Jhelum section are transformed to k_{max} axes in the eastern Neelum section. At the same time the horizontal bedding plane would rotate to a steep orientation, as observed in the field. The scatter of AMS ellipsoids in B (Neelum east) is about the same as that in A (Jhelum).

The transformation from the eastern Neelum (B) to the eastern Kaghan valley (C) is characterized by a change from flattened to constricted AMS ellipsoids. The principal anisotropy axes remain constant or change their orientation only slightly. Increasing shear strain on the eastern Neelum fabric (B), which is possibly accompanied by a negative volume change, would result in the eastern Kaghan fabric (C).

In the western Kaghan section (D) the AMS ellipsoids trend again toward the flattening field, with the principal anisotropy axes maintaining their orientation. Field mapping of reduction spots in the Kaghan valley has demonstrated the same behavior: a constricted field in the middle and eastern part of the section and a flattening field in the vicinity of the mylonite shear zone in the west. In terms of displacements at shear zone margins the western Kaghan Murree formation can be considered as a part of the mylonite shear zone margin, where N-S directed compression occurs [Bossart et al., 1988]. The D fabric could therefore be considered as a superposition of this compressive strain onto the C fabric, resulting in less constricted or slightly flattened ellipsoids.

The superposition of NE-SW and NNW-SSE directed compressive strains probably has changed the flattened sedimentary ellipsoids (A) to constricted ones with subvertical long axes in the western Neelum section (E) . This suggestion is supported by the occurrence of refolded folds which were mapped in the western Neelum valley at the base of the Murree formation. Thus two deformation paths from A to E (Figure 11b, right-hand side) are possible: a direct and a deflected deformation path. The latter deformation path is analogous to the path from A to B.

6. CONCLUSIONS

1. Characteristic remanence directions (ChRM) residing in hematite obtained for three sections of the Hazara-Kashmir Syntaxis were obtained by thermal cleaning. Polarity changes in the Jhelum section, which is only slightly deformed, give evidence for an early if not primary origin of the ChRM.

2. Dip-corrected inclination values of the Jhelum section suggest sedimentation at a paleolatitude of about 8°N. In comparison with the present day latitude of the Murree beds in the syntaxis, the Murree foreland basin has been displaced with the Indian craton at least 2600 km to the north.

3. Synthesis of structural and paleomagnetic data leads to a new kinematic model for the formation of the HKS. The early mass transport direction was initially opposite to the generally northward directed movement of India and was then passively rotated in a clockwise sense to its present position, which is directed toward the SW (220°). The late mass transport direction, which is observed in the southwest of the HKS, was always roughly antiparallel to the Indian plate motion. A slight counterclockwise rotation of the HKS relative to India to achieve its present SSE (160°) alignment is suggested by Tertiary paleomagnetic data from the Salt range and the Potwar Plateau.

4. In the Neelum and Kaghan sections the ChRM directions have been passively rotated into the cleavage plane. The ChRM directions are randomly oriented within the cleavage planes and are not aligned parallel to the maximum extension direction in these

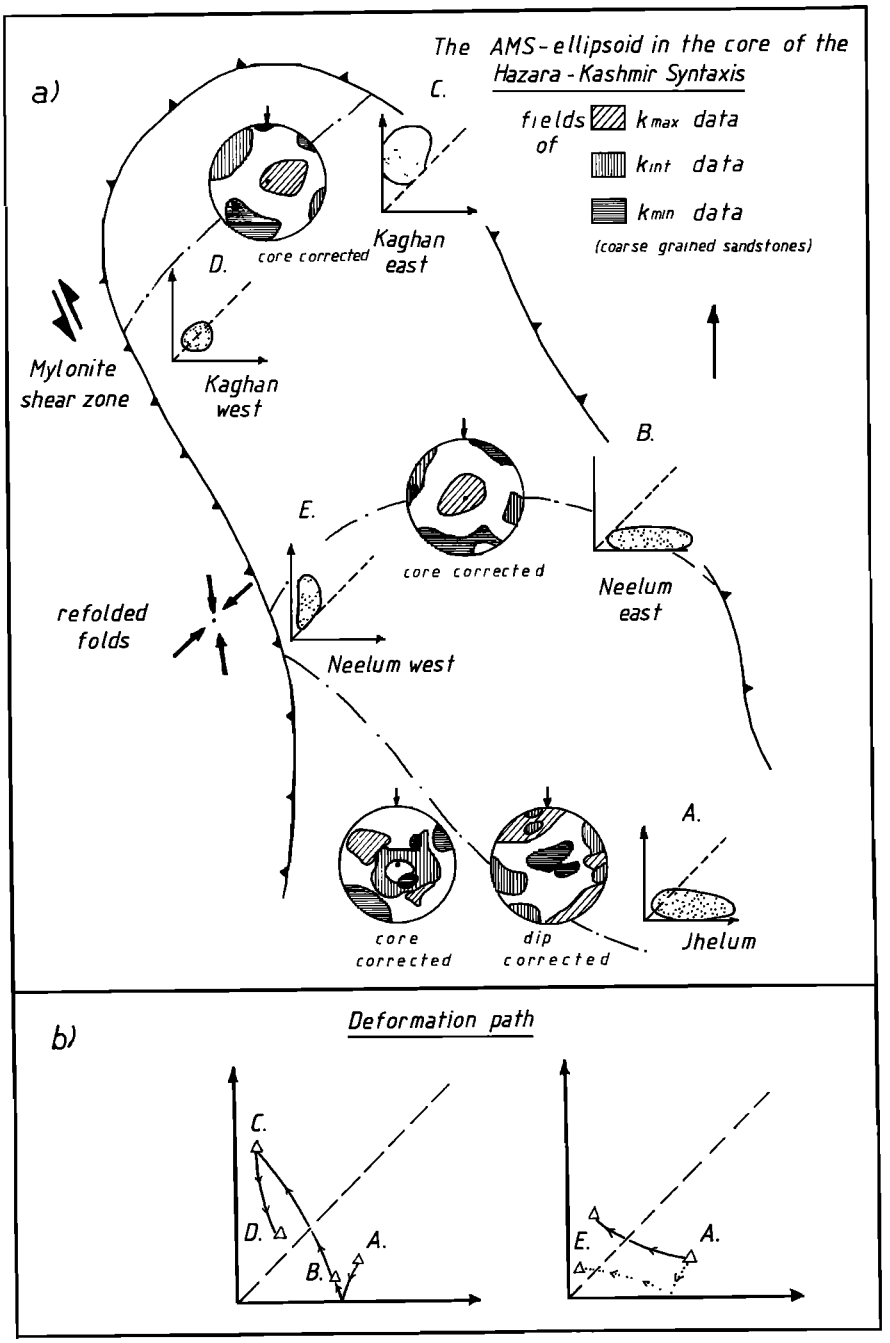


Fig. 11. (a) Development of AMS ellipsoids of sandstones in the core of the syntaxis. The influence of the mylonite shear zone and tectonic superposition (refolded folds) requires a division of the Flinn diagrams of the Neelum and Kaghan sections into an eastern and western part; the stereoplots show both parts of these sections. (b) A qualitative deformation path; left, a path from the Jhelum section (A) over the eastern Neelum (B) and eastern Kaghan (C) sections to the western Kaghan section (D); right, two alternative paths from the Jhelum section (A) to the western Neelum section (E).

two northern sections. The Neelum and Kaghan sections are not suitable for a regional paleomagnetic interpretation.

5. In the Jhelum section, where the Murree beds have been rigidly rotated without internal deformation, the orientation and shape of AMS ellipsoids reflect sedimentary compaction. The superposition of various tectonic strains (cleavage, mylonite shear zone, refolded folds) in the Neelum and Kaghan sections leads to orientations and shapes of AMS ellipsoids which may be correlated with the ellipsoidal forms of deformed reduction spots. A deformation path from south to north is suggested in the light of the deformation phases recorded previously in the syntaxis [Bossart et al., 1988]. Furthermore, the AMS data confirm the conclusion that the NRM measurements of the Jhelum section are reliable for a regional tectonic interpretation.

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